G. KARTHIK

Cybersecurity Student | Aspiring Penetration Tester
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PROFILE

Third-year Cybersecurity student at Shiv Nadar University Chennai with hands-on experience in offensive security, endpoint incident response, and threat detection engineering. Completed a summer internship in information security at Maveric Systems, working across vulnerability management, EDR triage, and enterprise security policy. Ranked Top 20% on TryHackMe ; actively developing detection rules, malware analysis tools, and penetration testing skills. Seeking offensive security roles where practical skills in VAPT, detection engineering, and scripting can be applied.

EDUCATION

B.Tech in Computer Science and Engineering (Cybersecurity) 2024 – 2028
Shiv Nadar University Chennai *3rd Year*

EXPERIENCE

Information Security Analyst Intern May 2026 – July 2026
Maveric Systems, Chennai

- Designed and implemented enterprise password security policies to strengthen authentication practices and align with organizational security standards.
- Investigated and triaged endpoint security incidents using **SentinelOne, Netskope, Tenable Nessus, and Trellix DLP** — analyzing malware alerts, suspicious executables, compromised credentials, and unauthorized software.
- Conducted threat research on OWASP Top 10, recent CVEs, Kerberos attacks, and emerging AI security risks; applied findings directly to active incident investigations and remediation workflows.
- Collaborated with security analysts and end users to validate alerts, perform root cause analysis, and recommend mitigation strategies across enterprise endpoints.

PROJECTS

WSL-Based EDR Evasion Detection Research
Stack: Sigma Rules, Sysmon, Windows Security, MITRE ATT&CK, Detection Engineering

- Researched Windows Subsystem for Linux (WSL) abuse techniques used by threat actors to evade endpoint detection and response (EDR) solutions.
- Developed a high-fidelity Sigma detection rule to identify suspicious WSL process execution using Sysmon Event ID 1 telemetry.
- Validated detection logic through attack simulation, mapped techniques to the MITRE ATT&CK framework, and documented testing evidence.
- Submitted the detection rule to the **SigmaHQ community**, demonstrating practical experience in threat detection engineering and SIEM rule development.

SIEM Dashboard for DevOps – Self-Hosted Security Monitoring Platform *Team of 4*
Python, FastAPI, PostgreSQL, OpenSearch, React, Docker Compose, MITRE ATT&CK

- Built an end-to-end SIEM that ingests Windows, Linux, Nginx, and Docker logs into one normalised schema, with alerting, incident workflow, GeoIP attack map, and CSV/PDF reports.
- Co-developed five detection rules (brute force, multiple failed logins, port scan, suspicious command, malware indicators) with correlation windows and alert suppression, mapped to MITRE ATT&CK techniques (T1110, T1046, T1059).
- Owned the FastAPI backend and PostgreSQL schema: ingestion API with key authentication and rate limiting, JWT authentication with Argon2id hashing, and role-based access control.
- Backend covered by 62 automated tests (91% coverage) and 13/13 end-to-end smoke checks; Trivy, gitleaks, and dependency audits returned 0 high/critical findings.

Magic Number Analyzer – Static Malware Analysis Tool
Stack: Python, hashlib, pathlib, File I/O, Static Analysis

- Built a static file analysis tool to detect malicious or disguised files by validating file signatures (magic numbers) against declared file extensions.
- Implemented SHA-256 hashing, entropy analysis, and rule-based risk assessment to identify packed, encrypted, or suspicious files.
- Automated detection of extension mismatches and generated forensic insights aligned with standard malware triage workflows.

Web Application VAPT – Vulnerability Assessment & Penetration Testing
Stack: Burp Suite, Wireshark, Kali Linux, OWASP Top 10

- Intercepted and modified live HTTP/HTTPS traffic using Burp Suite to identify input validation flaws and session handling weaknesses across multiple web application endpoints.
- Discovered and manually verified SQL Injection and Reflected XSS vulnerabilities; confirmed exploitability using custom payloads and documented proof-of-concept steps.
- Produced a structured vulnerability report with CVSS severity ratings and developer-facing remediation guidance for each identified finding.

Penetration Testing Home Lab
Stack: Metasploit, Kali Linux, VirtualBox, Nmap, Hydra

- Designed an isolated virtual network hosting Metasploitable 2, DVWA, and a Windows target to simulate realistic corporate attack surfaces.
- Conducted full-scope assessments including port scanning, service enumeration, exploitation via Metasploit, and post-exploitation privilege escalation on Linux targets.
- Practiced credential brute-forcing with Hydra; maintained structured notes following a professional penetration test report format covering scope, methodology, findings, and mitigations.

TECHNICAL SKILLS

- **Offensive Tools:** Burp Suite, Metasploit, Nmap, Gobuster, Hydra
- **Defensive & EDR Tools:** SentinelOne, Netskope, Tenable Nessus, Trellix DLP
- **Detection Engineering:** Sigma Rules, Sysmon, MITRE ATT&CK, SIEM Rule Development
- **Analysis & Monitoring:** Wireshark, TCP/IP, HTTP/HTTPS, Packet Analysis, Static Malware Analysis
- **Web Security:** SQL Injection (SQLi), XSS, Authentication Bypass, OWASP Top 10
- **Platforms:** Kali Linux, VirtualBox, VMware, Windows Security
- **Scripting:** Python, Bash
- **Core Competencies:** Vulnerability Assessment, Penetration Testing, Incident Triage, Threat Research, Report Writing

CERTIFICATIONS

- Junior Penetration Tester Path — TryHackMe, 2024
- Cybersecurity for Blockchain — EC-Council (Score: 90%)
- Advent of Cyber 2025 — TryHackMe, 2025

ACHIEVEMENTS

- Ranked in the **Top 20%** globally on TryHackMe out of over 7 million registered users.
- Completed **43+ hands-on labs** covering web exploitation, privilege escalation, and network security.
- Contributed a detection rule to the **SigmaHQ open-source community** for WSL-based EDR evasion techniques.

CURRENTLY BUILDING

- Preparing for the **eJPT** (eLearnSecurity Junior Penetration Tester) certification through structured lab practice.
- Actively solving machines on **Hack The Box** and participating in CTF competitions; documenting writeups on GitHub.